SQL Phase 2 Capstone Project Report

UK Air Quality Monitoring Sites - 500 Row Dataset

1. Project Overview

Item	Details
Domain	Environment and Climate - UK Air Quality Monitoring Coverage
Dataset	UK AIR Sites clean 500-row sample prepared for SQL Server
Dataset size	500 rows and 16 columns
Source / origin	Defra UK-AIR (UK Air Information Resource) - monitoring-site information (Search for monitoring sites): https://uk-air.defra.gov.uk/networks/find-sites?site_name=&pollutant=9999&group_id=9999&closed=true&country_id=9999&region_id=9999&location_type=9999&search=Search+Network&view=advanced&action=results. A cleaned 500-row sample of UK air quality monitoring-site metadata (site identifiers, environment type, zone, coordinates, monitoring start/end dates, networks and site descriptions). Source data is Crown copyright, published by Defra via uk-air.defra.gov.uk under the Open Government Licence v3.0.
End consumer	A UK local authority air quality and environmental monitoring manager who needs to identify monitoring coverage gaps, site-type patterns, lifecycle trends, network coverage and data-quality issues.
SQL platform used	Microsoft SQL Server / SSMS syntax. The main table is dbo.ShraddhaYadav_UK_AIR_Sites_500.

2. Why This Is a Social-Impact Project

Air quality monitoring is linked to public health, transport planning and environmental reporting. A local authority needs to know where monitoring sites exist, which areas are covered, what type of environments are monitored and whether the records have enough metadata to support decisions. This dataset supports that work because it contains monitoring site locations, site character, start and end dates, networks, identifiers and descriptions.

3. Five Analytical Perspectives

1. Geographic Coverage: Shows how the 500 monitoring sites are distributed across zones and helps identify areas with strong or weak coverage.

2. Environment Type and Site Character: Checks whether the monitoring network is mainly traffic-focused, background-focused, rural-focused or mixed.

3. Time and Monitoring Lifecycle: Explains when sites started and ended and highlights long-running or short-running monitoring locations.

4. Identifier, Network and Metadata Coverage: Reviews EU IDs, EMEP IDs, network values and descriptions so the manager can judge record completeness.

5. Operational Prioritisation and Data Quality: Combines site type, duration, altitude and metadata quality to support practical prioritisation.

4. Dataset Profile

Item	Details
Rows	500
Columns	16
Zones	35
Environment Types	9
Network Values	28
Records with EU Site ID	104
Records with EMEP Site ID	13

Records with Site Description	41
Coordinate Complete Records	500
Earliest Start Date	1961-04-04
Latest End Date	2025-01-09

5. SQL Concept Coverage

SQL Concept	Number of Queries
Window Function	14
GROUP BY / Aggregation	13
Basic WHERE	13
GROUP BY / CASE	6
CTE	5
ORDER BY	4
GROUP BY / HAVING	4
LIKE Filter	4
VIEW	3
DISTINCT / ORDER BY	2
Basic SELECT	1
Date Calculation / Aggregation	1
VIEW / Window Function	1
GROUP BY / Text Length	1
CASE Expression	1
CTE / CASE	1
CTE / Window Function	1

Sample shown: 17 of 17 returned row(s).

The project contains exactly 75 questions split into five perspectives. Each perspective has 15 questions: 5 easy, 5 medium and 5 difficult. The difficult questions include CTEs, views and window functions. The SQL file is organised in the same order so it can be followed from top to bottom.

6. Query Index, Sample Outputs and Interpretations

Geographic Coverage

Q01 - Geographic Coverage - Easy

Business question: List the first 20 monitoring sites with their zone, latitude, and longitude.

SQL concept: Basic SELECT

SQL Server query:

```
-- Q01 | Geographic Coverage | Easy | Basic SELECT
-- List the first 20 monitoring sites with their zone, latitude, and longitude.

SELECT TOP (20) id, uk_air_id, site_name, zone, latitude, longitude
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
ORDER BY id;
```

Sample output:

id	uk_air_id	site_name	zone	latitude	longitude
1	UKA15910	SH15 SH20 SH25 Verran Road	South East	51.3222	-0.743709
2	UKA15956	SH15 SH22 SH25 Air Quality Machine Bisley	South East	51.3209	-0.63089
3	UKA13340	ABERDARE 1N	South Wales	51.7123	-3.44734
4	UKA13341	ABERDARE 2N	South Wales	51.7124	-3.43721
5	UKA00399	Aberdeen	North East Scotland	57.1574	-2.09428

Sample shown: 5 of 20 returned row(s).

Interpretation: The query returns 20 row(s). This helps compare monitoring-site coverage across places and identify stronger or weaker geographic coverage.

Q02 - Geographic Coverage - Easy

Business question: Which sites are located in the South East zone?

SQL concept: Basic WHERE

SQL Server query:

```
-- Q02 | Geographic Coverage | Easy | Basic WHERE
-- Which sites are located in the South East zone?
```

```
SELECT uk_air_id, site_name, zone
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
WHERE zone = N'South East'
ORDER BY site_name;
```

Sample output:

uk_air_id	site_name	zone
UKA14669	ABINGDON 2N	South East
UKA13353	ADDLESTONE 2N	South East
UKA14529	ALTON 2N	South East
UKA14530	ALTON 4N	South East
UKA14531	ALTON 5N	South East

Sample shown: 5 of 56 returned row(s).

Interpretation: The query returns 56 row(s). This helps compare monitoring-site coverage across places and identify stronger or weaker geographic coverage.

Q03 - Geographic Coverage - Easy

Business question: Which unique UK zones are present in the 500-row dataset?

SQL concept: DISTINCT / ORDER BY

SQL Server query:

```
-- Q03 | Geographic Coverage | Easy | DISTINCT / ORDER BY
-- Which unique UK zones are present in the 500-row dataset?

SELECT DISTINCT zone
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
ORDER BY zone;
```

Sample output:

zone
Belfast Metropolitan Urban Area
Birkenhead Urban Area
Blackpool Urban Area
Brighton/Worthing/Littlehampton
Bristol Urban Area

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps compare monitoring-site coverage across places and identify stronger or weaker geographic coverage.

Q04 - Geographic Coverage - Easy

Business question: Which sites have valid latitude and longitude values?

SQL concept: Basic WHERE

SQL Server query:

```
-- Q04 | Geographic Coverage | Easy | Basic WHERE
-- Which sites have valid latitude and longitude values?

SELECT TOP (25) uk_air_id, site_name, zone, latitude, longitude
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
WHERE latitude IS NOT NULL AND longitude IS NOT NULL
ORDER BY zone, site_name;
```

Sample output:

uk_air_id	site_name	zone	latitude	longitude
UKA13419	BELFAST 2N	Belfast Metropolitan Urban Area	54.6127	-5.90899
UKA00371	Belfast Clara St	Belfast Metropolitan Urban Area	54.5913	-5.89546
UKA00175	Belfast East	Belfast Metropolitan Urban Area	54.5965	-5.90167
UKA00228	Belfast South	Belfast Metropolitan Urban Area	54.5992	-5.91242
UKA15206	CARRICKFERGUS 2N	Belfast Metropolitan Urban Area	55.116	-6.05847

Sample shown: 5 of 25 returned row(s).

Interpretation: The query returns 25 row(s). This helps compare monitoring-site coverage across places and identify stronger or weaker geographic coverage.

Q05 - Geographic Coverage - Easy

Business question: Which sites have altitude above 100 metres?

SQL concept: Basic WHERE

SQL Server query:

```
-- Q05 | Geographic Coverage | Easy | Basic WHERE
-- Which sites have altitude above 100 metres?

SELECT uk_air_id, site_name, zone, altitude_m
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
WHERE altitude_m > 100
ORDER BY altitude_m DESC;
```

Sample output:

uk_air_id	site_name	zone	altitude_m
UKA13727	DUNDEE 1N	North East Scotland	805
UKA00100	Cow Green Reservoir	North East	510
UKA00242	Bylchau	North Wales	390
UKA00120	Beddgelert	North Wales	358
UKA13535	BRYNMAWR 3N	South Wales	330

Sample shown: 5 of 110 returned row(s).

Interpretation: The query returns 110 row(s). This helps compare monitoring-site coverage across places and identify stronger or weaker geographic coverage.

Q06 - Geographic Coverage - Medium

Business question: How many monitoring sites are there in each zone?

SQL concept: GROUP BY / Aggregation

SQL Server query:

```
-- Q06 | Geographic Coverage | Medium | GROUP BY / Aggregation
-- How many monitoring sites are there in each zone?

SELECT zone, COUNT(*) AS total_sites
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY zone
ORDER BY total_sites DESC, zone;
```

Sample output:

zone	total_sites
South East	56
Northern Ireland	44
Eastern	40
South Wales	38
East Midlands	34

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps compare monitoring-site coverage across places and identify stronger or weaker geographic coverage.

Q07 - Geographic Coverage - Medium

Business question: What is the average altitude of monitoring sites in each zone?

SQL concept: GROUP BY / Aggregation

SQL Server query:

```
-- Q07 | Geographic Coverage | Medium | GROUP BY / Aggregation
-- What is the average altitude of monitoring sites in each zone?

SELECT zone, ROUND(AVG(CAST(altitude_m AS FLOAT)), 2) AS avg_altitude_m, COUNT(*) AS site_count
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
```

```
GROUP BY zone
ORDER BY avg_altitude_m DESC;
```

Sample output:

zone	avg_altitude_m	site_count
Preston Urban Area	267	2
North Wales	200	4
Scottish Borders	185	3
North East Scotland	157.67	9
West Midlands Urban Area	133.19	21

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps compare monitoring-site coverage across places and identify stronger or weaker geographic coverage.

Q08 - Geographic Coverage - Medium

Business question: Which zones have at least 10 monitoring sites?

SQL concept: GROUP BY / HAVING

SQL Server query:

```
-- Q08 | Geographic Coverage | Medium | GROUP BY / HAVING
-- Which zones have at least 10 monitoring sites?

SELECT zone, COUNT(*) AS total_sites
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY zone
HAVING COUNT(*) >= 10
ORDER BY total_sites DESC, zone;
```

Sample output:

zone	total_sites
South East	56
Northern Ireland	44
Eastern	40
South Wales	38
East Midlands	34

Sample shown: 5 of 14 returned row(s).

Interpretation: The query returns 14 row(s). This helps compare monitoring-site coverage across places and identify stronger or weaker geographic coverage.

Q09 - Geographic Coverage - Medium

Business question: How many mapped and unmapped sites exist in each zone?

SQL concept: GROUP BY / CASE

SQL Server query:

```
-- Q09 | Geographic Coverage | Medium | GROUP BY / CASE
-- How many mapped and unmapped sites exist in each zone?

SELECT zone,
       CASE WHEN latitude IS NOT NULL AND longitude IS NOT NULL THEN 'Coordinates Available' ELSE
    'Coordinates Missing' END AS coordinate_status,
       COUNT(*) AS site_count
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY zone, CASE WHEN latitude IS NOT NULL AND longitude IS NOT NULL THEN 'Coordinates Available' ELSE
    'Coordinates Missing' END
ORDER BY zone, coordinate_status;
```

Sample output:

zone	coordinate_status	site_count
Belfast Metropolitan Urban Area	Coordinates Available	6
Birkenhead Urban Area	Coordinates Available	7
Blackpool Urban Area	Coordinates Available	6
Brighton/Worthing/Littlehampton	Coordinates Available	3
Bristol Urban Area	Coordinates Available	7

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps compare monitoring-site coverage across places and identify stronger or weaker geographic coverage.

Q10 - Geographic Coverage - Medium

Business question: Which zone and environment type combinations have the most sites?

SQL concept: GROUP BY / Aggregation

SQL Server query:

```
-- Q10 | Geographic Coverage | Medium | GROUP BY / Aggregation
-- Which zone and environment type combinations have the most sites?

SELECT TOP (15) zone, environment_type, COUNT(*) AS total_sites
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY zone, environment_type
ORDER BY total_sites DESC, zone, environment_type;
```

Sample output:

zone	environment_type	total_sites
South East	Urban Traffic	47
South Wales	Urban Traffic	35
Eastern	Urban Traffic	34
Greater London Urban Area	Urban Traffic	31
Central Scotland	Urban Traffic	28

Sample shown: 5 of 15 returned row(s).

Interpretation: The query returns 15 row(s). This helps compare monitoring-site coverage across places and identify stronger or weaker geographic coverage.

Q11 - Geographic Coverage - Difficult

Business question: Rank zones by total number of monitoring sites.

SQL concept: Window Function

SQL Server query:

```
-- Q11 | Geographic Coverage | Difficult | Window Function
-- Rank zones by total number of monitoring sites.

WITH zone_counts AS (
    SELECT zone, COUNT(*) AS total_sites
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
    GROUP BY zone
)
SELECT zone, total_sites, RANK() OVER (ORDER BY total_sites DESC) AS coverage_rank
FROM zone_counts
ORDER BY coverage_rank, zone;
```

Sample output:

zone	total_sites	coverage_rank
South East	56	1
Northern Ireland	44	2
Eastern	40	3
South Wales	38	4
East Midlands	34	5

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps compare monitoring-site coverage across places and identify stronger or weaker geographic coverage.

Q12 - Geographic Coverage - Difficult

Business question: Show each zone count and its percentage of all monitoring sites.

SQL concept: Window Function

SQL Server query:

```
-- Q12 | Geographic Coverage | Difficult | Window Function
-- Show each zone count and its percentage of all monitoring sites.

WITH zone_counts AS (
    SELECT zone, COUNT(*) AS total_sites
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
    GROUP BY zone
)
SELECT zone, total_sites,
       ROUND(100.0 * total_sites / SUM(total_sites) OVER (), 2) AS pct_of_all_sites
FROM zone_counts
ORDER BY total_sites DESC, zone;
```

Sample output:

zone	total_sites	pct_of_all_sites
South East	56	11.2
Northern Ireland	44	8.8
Eastern	40	8
South Wales	38	7.6
East Midlands	34	6.8

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps compare monitoring-site coverage across places and identify stronger or weaker geographic coverage.

Q13 - Geographic Coverage - Difficult

Business question: Use a CTE to identify zones with below-average site coverage.

SQL concept: CTE

SQL Server query:

```
-- Q13 | Geographic Coverage | Difficult | CTE
-- Use a CTE to identify zones with below-average site coverage.

WITH zone_counts AS (
    SELECT zone, COUNT(*) AS total_sites
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
    GROUP BY zone
), overall_avg AS (
    SELECT AVG(CAST(total_sites AS FLOAT)) AS avg_sites_per_zone
    FROM zone_counts
)
SELECT z.zone, z.total_sites, ROUND(o.avg_sites_per_zone, 2) AS average_zone_sites
FROM zone_counts z
CROSS JOIN overall_avg o
WHERE z.total_sites < o.avg_sites_per_zone
ORDER BY z.total_sites ASC, z.zone;
```

Sample output:

zone	total_sites	average_zone_sites
Not Available	1	14.29
Teesside Urban Area	1	14.29
Liverpool Urban Area	2	14.29
Preston Urban Area	2	14.29
Reading/Wokingham Urban Area	2	14.29

Sample shown: 5 of 22 returned row(s).

Interpretation: The query returns 22 row(s). This helps compare monitoring-site coverage across places and identify stronger or weaker geographic coverage.

Q14 - Geographic Coverage - Difficult

Business question: Use the zone summary view to review overall geographic coverage.

SQL concept: VIEW

SQL Server query:

```
-- Q14 | Geographic Coverage | Difficult | VIEW
-- Use the zone summary view to review overall geographic coverage.

SELECT zone, total_sites, environment_type_count, avg_altitude_m, first_start_date, last_end_date,
    avg_operational_years
```

```
FROM dbo.vw_zone_summary_500
ORDER BY total_sites DESC, zone;
```

Sample output:

zone	total_sites	environment_type_count	avg_altitude_m	first_start_date	last_end_date	avg_operational_years
South East	56	4	59.98	1975-04-18	2024-11-04	9.25
Northern Ireland	44	5	51.64	1986-01-07	2024-09-03	5.35
Eastern	40	3	35.33	1983-03-15	2023-10-11	8.66
South Wales	38	3	81.13	1991-08-13	2020-08-07	8.54
East Midlands	34	3	82.71	1977-10-01	2016-10-17	8.45

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps compare monitoring-site coverage across places and identify stronger or weaker geographic coverage.

Q15 - Geographic Coverage - Difficult

Business question: Find the northernmost monitoring site in each zone using row numbering.

SQL concept: Window Function

SQL Server query:

```
-- Q15 | Geographic Coverage | Difficult | Window Function
-- Find the northernmost monitoring site in each zone using row numbering.

WITH ranked_sites AS (
    SELECT zone, uk_air_id, site_name, latitude,
           ROW_NUMBER() OVER (PARTITION BY zone ORDER BY latitude DESC, site_name) AS rn
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
)
SELECT zone, uk_air_id, site_name, latitude
FROM ranked_sites
WHERE rn = 1
ORDER BY zone;
```

Sample output:

zone	uk_air_id	site_name	latitude
Belfast Metropolitan Urban Area	UKA15206	CARRICKFERGUS 2N	55.116
Birkenhead Urban Area	UKA15007	BIRKENHEAD 1N	53.3988
Blackpool Urban Area	UKA13449	BLACKPOOL 2N	53.823
Brighton/Worthing/Littlehampton	UKA13509	BRIGHTON 3N	50.8333
Bristol Urban Area	UKA00127	Avonmouth 2	51.5054

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps compare monitoring-site coverage across places and identify stronger or weaker geographic coverage.

Environment Type and Site Character

Q16 - Environment Type and Site Character - Easy

Business question: List sites classified as Urban Traffic.

SQL concept: Basic WHERE

SQL Server query:

```
-- Q16 | Environment Type and Site Character | Easy | Basic WHERE
-- List sites classified as Urban Traffic.

SELECT TOP (30) uk_air_id, site_name, zone, environment_type
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
WHERE environment_type = N'Urban Traffic'
ORDER BY zone, site_name;
```

Sample output:

uk_air_id	site_name	zone	environment_type
UKA13419	BELFAST 2N	Belfast Metropolitan Urban Area	Urban Traffic
UKA15206	CARRICKFERGUS 2N	Belfast Metropolitan Urban Area	Urban Traffic
UKA15007	BIRKENHEAD 1N	Birkenhead Urban Area	Urban Traffic
UKA13438	BIRKENHEAD 2N	Birkenhead Urban Area	Urban Traffic
UKA13757	EASTHAM WIRRAL 1N	Birkenhead Urban Area	Urban Traffic

Sample shown: 5 of 30 returned row(s).

Interpretation: The query returns 30 row(s). This helps compare site character, such as traffic, background and rural monitoring environments.

Q17 - Environment Type and Site Character - Easy

Business question: List sites classified as Rural Background.

SQL concept: Basic WHERE

SQL Server query:

```
-- Q17 | Environment Type and Site Character | Easy | Basic WHERE
-- List sites classified as Rural Background.

SELECT uk_air_id, site_name, zone, environment_type
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
WHERE environment_type = N'Rural Background'
ORDER BY zone, site_name;
```

Sample output:

uk_air_id	site_name	zone	environment_type
UKA00202	Camphill 1	Central Scotland	Rural Background
UKA00464	Beacon Hill	East Midlands	Rural Background
UKA00055	Bottesford	East Midlands	Rural Background
UKA00190	Caenby 1	East Midlands	Rural Background
UKA00101	Driby	East Midlands	Rural Background

Sample shown: 5 of 58 returned row(s).

Interpretation: The query returns 58 row(s). This helps compare site character, such as traffic, background and rural monitoring environments.

Q18 - Environment Type and Site Character - Easy

Business question: Show all distinct environment types in the dataset.

SQL concept: DISTINCT / ORDER BY

SQL Server query:

```
-- Q18 | Environment Type and Site Character | Easy | DISTINCT / ORDER BY
-- Show all distinct environment types in the dataset.

SELECT DISTINCT environment_type
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
ORDER BY environment_type;
```

Sample output:

environment_type
Roadside
Rural Background
Suburban
Suburban Background
Suburban Industrial

Sample shown: 5 of 9 returned row(s).

Interpretation: The query returns 9 row(s). This helps compare site character, such as traffic, background and rural monitoring environments.

Q19 - Environment Type and Site Character - Easy

Business question: Find site names containing the word London.

SQL concept: LIKE Filter

SQL Server query:

```
-- Q19 | Environment Type and Site Character | Easy | LIKE Filter
-- Find site names containing the word London.
```

```
SELECT uk_air_id, site_name, zone, environment_type
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
WHERE site_name LIKE N'%London%'
ORDER BY site_name;
```

Sample output:

uk_air_id	site_name	zone	environment_type
UKA00009	Central London	Greater London Urban Area	Urban Background
UKA15819	DA31 Taunton Rd (off London Rd)	Greater London Urban Area	Urban Traffic

Sample shown: 2 of 2 returned row(s).

Interpretation: The query returns 2 row(s). This helps compare site character, such as traffic, background and rural monitoring environments.

Q20 - Environment Type and Site Character - Easy

Business question: Which sites are classified as a background environment type?

SQL concept: LIKE Filter

SQL Server query:

```
-- Q20 | Environment Type and Site Character | Easy | LIKE Filter
-- Which sites are classified as a background environment type?

SELECT uk_air_id, site_name, zone, environment_type
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
WHERE environment_type LIKE N'%Background%'
ORDER BY zone, site_name;
```

Sample output:

uk_air_id	site_name	zone	environment_type
UKA00371	Belfast Clara St	Belfast Metropolitan Urban Area	Suburban Background
UKA00175	Belfast East	Belfast Metropolitan Urban Area	Urban Background
UKA00228	Belfast South	Belfast Metropolitan Urban Area	Urban Background
UKA00412	Blackpool	Blackpool Urban Area	Urban Background
UKA00221	Bristol Centre	Bristol Urban Area	Urban Background

Sample shown: 5 of 92 returned row(s).

Interpretation: The query returns 92 row(s). This helps compare site character, such as traffic, background and rural monitoring environments.

Q21 - Environment Type and Site Character - Medium

Business question: How many sites exist for each environment type?

SQL concept: GROUP BY / Aggregation

SQL Server query:

```
-- Q21 | Environment Type and Site Character | Medium | GROUP BY / Aggregation
-- How many sites exist for each environment type?

SELECT environment_type, COUNT(*) AS total_sites
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY environment_type
ORDER BY total_sites DESC, environment_type;
```

Sample output:

environment_type	total_sites
Urban Traffic	392
Rural Background	58
Urban Background	30
Suburban	10
Suburban Background	4

Sample shown: 5 of 9 returned row(s).

Interpretation: The query returns 9 row(s). This helps compare site character, such as traffic, background and rural monitoring environments.

Q22 - Environment Type and Site Character - Medium

Business question: How many traffic-related and non-traffic sites exist?

SQL concept: GROUP BY / CASE

SQL Server query:

```
-- Q22 | Environment Type and Site Character | Medium | GROUP BY / CASE
-- How many traffic-related and non-traffic sites exist?

SELECT CASE WHEN environment_type LIKE N'%Traffic%' OR environment_type = N'Roadside' THEN 'Traffic /
    Roadside' ELSE 'Background / Other' END AS site_category,
       COUNT(*) AS total_sites
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY CASE WHEN environment_type LIKE N'%Traffic%' OR environment_type = N'Roadside' THEN 'Traffic /
    Roadside' ELSE 'Background / Other' END
ORDER BY total_sites DESC;
```

Sample output:

site_category	total_sites
Traffic / Roadside	393
Background / Other	107

Sample shown: 2 of 2 returned row(s).

Interpretation: The query returns 2 row(s). This helps compare site character, such as traffic, background and rural monitoring environments.

Q23 - Environment Type and Site Character - Medium

Business question: What is the average altitude by environment type?

SQL concept: GROUP BY / Aggregation

SQL Server query:

```
-- Q23 | Environment Type and Site Character | Medium | GROUP BY / Aggregation
-- What is the average altitude by environment type?

SELECT environment_type, ROUND(AVG(CAST(altitude_m AS FLOAT)), 2) AS avg_altitude_m, COUNT(*) AS site_count
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY environment_type
ORDER BY avg_altitude_m DESC;
```

Sample output:

environment_type	avg_altitude_m	site_count
Suburban Industrial	128	1
Rural Background	127.1	58
Urban Centre	100	1
Suburban Background	86.25	4
Urban Background	65.4	30

Sample shown: 5 of 9 returned row(s).

Interpretation: The query returns 9 row(s). This helps compare site character, such as traffic, background and rural monitoring environments.

Q24 - Environment Type and Site Character - Medium

Business question: Which environment types appear in more than one zone?

SQL concept: GROUP BY / HAVING

SQL Server query:

```
-- Q24 | Environment Type and Site Character | Medium | GROUP BY / HAVING
-- Which environment types appear in more than one zone?

SELECT environment_type, COUNT(DISTINCT zone) AS zone_count
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY environment_type
HAVING COUNT(DISTINCT zone) > 1
ORDER BY zone_count DESC, environment_type;
```

Sample output:

environment_type	zone_count
Urban Traffic	32
Urban Background	17
Rural Background	15
Suburban	4
Suburban Background	3

Sample shown: 5 of 6 returned row(s).

Interpretation: The query returns 6 row(s). This helps compare site character, such as traffic, background and rural monitoring environments.

Q25 - Environment Type and Site Character - Medium

Business question: Which environment type and zone combinations have the highest average altitude?

SQL concept: GROUP BY / Aggregation

SQL Server query:

```
-- Q25 | Environment Type and Site Character | Medium | GROUP BY / Aggregation
-- Which environment type and zone combinations have the highest average altitude?

SELECT TOP (15) environment_type, zone, ROUND(AVG(CAST(altitude_m AS FLOAT)), 2) AS avg_altitude_m, COUNT(*)
    AS site_count
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY environment_type, zone
ORDER BY avg_altitude_m DESC, site_count DESC;
```

Sample output:

environment_type	zone	avg_altitude_m	site_count
Rural Background	North East	510	1
Rural Background	North Wales	374	2
Urban Traffic	Preston Urban Area	267	2
Urban Traffic	North East Scotland	247.75	4
Rural Background	North West & Merseyside	224.4	5

Sample shown: 5 of 15 returned row(s).

Interpretation: The query returns 15 row(s). This helps compare site character, such as traffic, background and rural monitoring environments.

Q26 - Environment Type and Site Character - Difficult

Business question: Rank environment types by total site coverage.

SQL concept: Window Function

SQL Server query:

```
-- Q26 | Environment Type and Site Character | Difficult | Window Function
-- Rank environment types by total site coverage.

WITH env_counts AS (
    SELECT environment_type, COUNT(*) AS total_sites
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
    GROUP BY environment_type
)
SELECT environment_type, total_sites, RANK() OVER (ORDER BY total_sites DESC) AS environment_rank
FROM env_counts
ORDER BY environment_rank, environment_type;
```

Sample output:

environment_type	total_sites	environment_rank
Urban Traffic	392	1
Rural Background	58	2
Urban Background	30	3
Suburban	10	4
Suburban Background	4	5

Sample shown: 5 of 9 returned row(s).

Interpretation: The query returns 9 row(s). This helps compare site character, such as traffic, background and rural monitoring environments.

Q27 - Environment Type and Site Character - Difficult

Business question: Compare each environment type average altitude against the overall average altitude.

SQL concept: CTE

SQL Server query:

```
-- Q27 | Environment Type and Site Character | Difficult | CTE
-- Compare each environment type average altitude against the overall average altitude.

WITH env_altitude AS (
    SELECT environment_type, ROUND(AVG(CAST(altitude_m AS FLOAT)), 2) AS avg_altitude_m
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
    GROUP BY environment_type
), overall_altitude AS (
    SELECT ROUND(AVG(CAST(altitude_m AS FLOAT)), 2) AS overall_avg_altitude
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
)
SELECT e.environment_type, e.avg_altitude_m, o.overall_avg_altitude,
       CASE WHEN e.avg_altitude_m > o.overall_avg_altitude THEN 'Above overall average' ELSE 'At or below
    overall average' END AS altitude_comparison
FROM env_altitude e
CROSS JOIN overall_altitude o
ORDER BY e.avg_altitude_m DESC;
```

Sample output:

environment_type	avg_altitude_m	overall_avg_altitude	altitude_comparison
Suburban Industrial	128	66.88	Above overall average
Rural Background	127.1	66.88	Above overall average
Urban Centre	100	66.88	Above overall average
Suburban Background	86.25	66.88	Above overall average
Urban Background	65.4	66.88	At or below overall average

Sample shown: 5 of 9 returned row(s).

Interpretation: The query returns 9 row(s). This helps compare site character, such as traffic, background and rural monitoring environments.

Q28 - Environment Type and Site Character - Difficult

Business question: Use the environment summary view to compare environment types.

SQL concept: VIEW

SQL Server query:

```
-- Q28 | Environment Type and Site Character | Difficult | VIEW
-- Use the environment summary view to compare environment types.

SELECT environment_type, total_sites, zones_covered, avg_altitude_m, avg_operational_years
FROM dbo.vw_environment_summary_500
ORDER BY total_sites DESC, environment_type;
```

Sample output:

environment_type	total_sites	zones_covered	avg_altitude_m	avg_operational_years
Urban Traffic	392	32	58.97	7.32
Rural Background	58	15	127.1	11.61
Urban Background	30	17	65.4	11.16
Suburban	10	4	35	11.69
Suburban Background	4	3	86.25	3.82

Sample shown: 5 of 9 returned row(s).

Interpretation: The query returns 9 row(s). This helps compare site character, such as traffic, background and rural monitoring environments.

Q29 - Environment Type and Site Character - Difficult

Business question: Find the top zone for each environment type.

SQL concept: Window Function

SQL Server query:

```
-- Q29 | Environment Type and Site Character | Difficult | Window Function
-- Find the top zone for each environment type.

WITH env_zone_counts AS (
    SELECT environment_type, zone, COUNT(*) AS site_count
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
    GROUP BY environment_type, zone
), ranked AS (
    SELECT environment_type, zone, site_count,
           ROW_NUMBER() OVER (PARTITION BY environment_type ORDER BY site_count DESC, zone) AS rn
    FROM env_zone_counts
)
SELECT environment_type, zone, site_count
FROM ranked
WHERE rn = 1
ORDER BY environment_type;
```

Sample output:

environment_type	zone	site_count
Roadside	Northern Ireland	1
Rural Background	Highland	10
Suburban	Northern Ireland	7
Suburban Background	South East	2
Suburban Industrial	West Midlands Urban Area	1

Sample shown: 5 of 9 returned row(s).

Interpretation: The query returns 9 row(s). This helps compare site character, such as traffic, background and rural monitoring environments.

Q30 - Environment Type and Site Character - Difficult

Business question: Calculate the percentage of all sites represented by each environment type.

SQL concept: Window Function

SQL Server query:

```
-- Q30 | Environment Type and Site Character | Difficult | Window Function
-- Calculate the percentage of all sites represented by each environment type.

SELECT environment_type, COUNT(*) AS total_sites,
       ROUND(100.0 * COUNT(*) / SUM(COUNT(*)) OVER (), 2) AS pct_of_all_sites
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY environment_type
ORDER BY total_sites DESC, environment_type;
```

Sample output:

environment_type	total_sites	pct_of_all_sites
Urban Traffic	392	78.4
Rural Background	58	11.6
Urban Background	30	6
Suburban	10	2
Suburban Background	4	0.8

Sample shown: 5 of 9 returned row(s).

Interpretation: The query returns 9 row(s). This helps compare site character, such as traffic, background and rural monitoring environments.

Time and Monitoring Lifecycle

Q31 - Time and Monitoring Lifecycle - Easy

Business question: Which sites started before 1980?

SQL concept: Basic WHERE

SQL Server query:

```
-- Q31 | Time and Monitoring Lifecycle | Easy | Basic WHERE
-- Which sites started before 1980?

SELECT uk_air_id, site_name, zone, start_date
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
```

```
WHERE start_date < '1980-01-01'
ORDER BY start_date, site_name;
```

Sample output:

uk_air_id	site_name	zone	start_date
UKA00001	Bradford 6	West Yorkshire Urban Area	1961-04-04
UKA00002	Cardiff 12	Cardiff Urban Area	1961-04-04
UKA00009	Central London	Greater London Urban Area	1972-07-01
UKA00013	Cardiff Kerbside	Cardiff Urban Area	1973-07-28
UKA00031	Chilworth	South East	1975-04-18

Sample shown: 5 of 7 returned row(s).

Interpretation: The query returns 7 row(s). This helps explain how monitoring coverage changed over time and which sites operated for longer periods.

Q32 - Time and Monitoring Lifecycle - Easy

Business question: Which sites started in or after 2000?

SQL concept: Basic WHERE

SQL Server query:

```
-- Q32 | Time and Monitoring Lifecycle | Easy | Basic WHERE
-- Which sites started in or after 2000?

SELECT uk_air_id, site_name, zone, start_date
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
WHERE start_date >= '2000-01-01'
ORDER BY start_date, site_name;
```

Sample output:

uk_air_id	site_name	zone	start_date
UKA15285	CHICHESTER 10N	South East	2000-03-02
UKA15286	CHICHESTER 11N	South East	2000-03-02
UKA15287	CHICHESTER 12N	South East	2000-03-02
UKA15288	CHICHESTER 13N	South East	2000-03-02
UKA15289	CHICHESTER 9N	South East	2000-03-02

Sample shown: 5 of 90 returned row(s).

Interpretation: The query returns 90 row(s). This helps explain how monitoring coverage changed over time and which sites operated for longer periods.

Q33 - Time and Monitoring Lifecycle - Easy

Business question: Which sites ended before 2000?

SQL concept: Basic WHERE

SQL Server query:

```
-- Q33 | Time and Monitoring Lifecycle | Easy | Basic WHERE
-- Which sites ended before 2000?

SELECT uk_air_id, site_name, zone, end_date
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
WHERE end_date < '2000-01-01'
ORDER BY end_date, site_name;
```

Sample output:

uk_air_id	site_name	zone	end_date
UKA00032	East Kilbride	Central Scotland	1975-10-09
UKA00031	Chilworth	South East	1975-10-10
UKA00013	Cardiff Kerbside	Cardiff Urban Area	1976-11-23
UKA00074	Elkington 1	West Midlands Urban Area	1985-11-26
UKA00075	Elkington 2	West Midlands	1985-11-26

Sample shown: 5 of 112 returned row(s).

Interpretation: The query returns 112 row(s). This helps explain how monitoring coverage changed over time and which sites operated for longer periods.

Q34 - Time and Monitoring Lifecycle - Easy

Business question: Show the 20 earliest monitoring start dates.

SQL concept: ORDER BY

SQL Server query:

```
-- Q34 | Time and Monitoring Lifecycle | Easy | ORDER BY
-- Show the 20 earliest monitoring start dates.

SELECT TOP (20) uk_air_id, site_name, zone, start_date
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
ORDER BY start_date ASC, site_name;
```

Sample output:

uk_air_id	site_name	zone	start_date
UKA00001	Bradford 6	West Yorkshire Urban Area	1961-04-04
UKA00002	Cardiff 12	Cardiff Urban Area	1961-04-04
UKA00009	Central London	Greater London Urban Area	1972-07-01
UKA00013	Cardiff Kerbside	Cardiff Urban Area	1973-07-28
UKA00031	Chilworth	South East	1975-04-18

Sample shown: 5 of 20 returned row(s).

Interpretation: The query returns 20 row(s). This helps explain how monitoring coverage changed over time and which sites operated for longer periods.

Q35 - Time and Monitoring Lifecycle - Easy

Business question: Show the 20 latest monitoring end dates.

SQL concept: ORDER BY

SQL Server query:

```
-- Q35 | Time and Monitoring Lifecycle | Easy | ORDER BY
-- Show the 20 latest monitoring end dates.

SELECT TOP (20) uk_air_id, site_name, zone, end_date
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
ORDER BY end_date DESC, site_name;
```

Sample output:

uk_air_id	site_name	zone	end_date
UKA00239	Balquhidder 2	North East Scotland	2025-01-09
UKA00936	Bannisdale Beck	North West & Merseyside	2025-01-03
UKA00513	Aberdeen Union Street Roadside	North East Scotland	2024-12-31
UKA00424	Canterbury	South East	2024-11-04
UKA00949	Ballylinney Church	Northern Ireland	2024-09-03

Sample shown: 5 of 20 returned row(s).

Interpretation: The query returns 20 row(s). This helps explain how monitoring coverage changed over time and which sites operated for longer periods.

Q36 - Time and Monitoring Lifecycle - Medium

Business question: How many sites started in each year?

SQL concept: GROUP BY / Aggregation

SQL Server query:

```
-- Q36 | Time and Monitoring Lifecycle | Medium | GROUP BY / Aggregation
-- How many sites started in each year?

SELECT YEAR(start_date) AS start_year, COUNT(*) AS sites_started
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY YEAR(start_date)
ORDER BY start_year;
```

Sample output:

start_year	sites_started
1961	2
1972	1

1973	1
1975	2
1977	1

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps explain how monitoring coverage changed over time and which sites operated for longer periods.

Q37 - Time and Monitoring Lifecycle - Medium

Business question: How many sites ended in each year?

SQL concept: GROUP BY / Aggregation

SQL Server query:

```
-- Q37 | Time and Monitoring Lifecycle | Medium | GROUP BY / Aggregation
-- How many sites ended in each year?

SELECT YEAR(end_date) AS end_year, COUNT(*) AS sites_ended
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY YEAR(end_date)
ORDER BY end_year;
```

Sample output:

end_year	sites_ended
1975	2
1976	1
1985	3
1986	2
1987	9

Sample shown: 5 of 41 returned row(s).

Interpretation: The query returns 41 row(s). This helps explain how monitoring coverage changed over time and which sites operated for longer periods.

Q38 - Time and Monitoring Lifecycle - Medium

Business question: What are the earliest and latest start dates by zone?

SQL concept: GROUP BY / Aggregation

SQL Server query:

```
-- Q38 | Time and Monitoring Lifecycle | Medium | GROUP BY / Aggregation
-- What are the earliest and latest start dates by zone?

SELECT zone, MIN(start_date) AS earliest_start_date, MAX(start_date) AS latest_start_date, COUNT(*) AS
    total_sites
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY zone
ORDER BY earliest_start_date, zone;
```

Sample output:

zone	earliest_start_date	latest_start_date	total_sites
Cardiff Urban Area	1961-04-04	2008-01-01	5
West Yorkshire Urban Area	1961-04-04	1997-11-28	2
Greater London Urban Area	1972-07-01	2005-11-29	32
South East	1975-04-18	2008-01-01	56
Central Scotland	1975-05-08	2001-01-02	30

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps explain how monitoring coverage changed over time and which sites operated for longer periods.

Q39 - Time and Monitoring Lifecycle - Medium

Business question: Which start years have at least 10 new monitoring sites?

SQL concept: GROUP BY / HAVING

SQL Server query:

```
-- Q39 | Time and Monitoring Lifecycle | Medium | GROUP BY / HAVING
-- Which start years have at least 10 new monitoring sites?

SELECT YEAR(start_date) AS start_year, COUNT(*) AS sites_started
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY YEAR(start_date)
HAVING COUNT(*) >= 10
ORDER BY sites_started DESC, start_year;
```

Sample output:

start_year	sites_started
1992	239
2001	29
1986	28
1993	26
1996	19

Sample shown: 5 of 13 returned row(s).

Interpretation: The query returns 13 row(s). This helps explain how monitoring coverage changed over time and which sites operated for longer periods.

Q40 - Time and Monitoring Lifecycle - Medium

Business question: What is the average operational duration for sites in each zone?

SQL concept: Date Calculation / Aggregation

SQL Server query:

```
-- Q40 | Time and Monitoring Lifecycle | Medium | Date Calculation / Aggregation
-- What is the average operational duration for sites in each zone?

SELECT zone, ROUND(AVG(CAST(DATEDIFF(DAY, start_date, end_date) AS FLOAT) / 365.25), 2) AS avg_years_operated,
    COUNT(*) AS site_count
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY zone
ORDER BY avg_years_operated DESC, zone;
```

Sample output:

zone	avg_years_operated	site_count
Teesside Urban Area	37.46	1
West Yorkshire Urban Area	29.42	2
Cardiff Urban Area	16.24	5
North East Scotland	14.42	9
Blackpool Urban Area	12.03	6

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps explain how monitoring coverage changed over time and which sites operated for longer periods.

Q41 - Time and Monitoring Lifecycle - Difficult

Business question: Use a CTE to classify site duration bands and count them by zone.

SQL concept: CTE

SQL Server query:

```
-- Q41 | Time and Monitoring Lifecycle | Difficult | CTE
-- Use a CTE to classify site duration bands and count them by zone.

WITH duration_cte AS (
    SELECT zone,
           CASE
               WHEN DATEDIFF(DAY, start_date, end_date) >= 3650 THEN '10 years or more'
               WHEN DATEDIFF(DAY, start_date, end_date) >= 1095 THEN '3 to 9 years'
               ELSE 'Under 3 years'
           END AS duration_band
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
)
SELECT zone, duration_band, COUNT(*) AS site_count
FROM duration_cte
GROUP BY zone, duration_band
ORDER BY zone, duration_band;
```

Sample output:

zone	duration_band	site_count
Belfast Metropolitan Urban Area	10 years or more	1
Belfast Metropolitan Urban Area	3 to 9 years	4
Belfast Metropolitan Urban Area	Under 3 years	1
Birkenhead Urban Area	10 years or more	3
Birkenhead Urban Area	3 to 9 years	3

Sample shown: 5 of 87 returned row(s).

Interpretation: The query returns 87 row(s). This helps explain how monitoring coverage changed over time and which sites operated for longer periods.

Q42 - Time and Monitoring Lifecycle - Difficult

Business question: Calculate year-over-year change in sites started.

SQL concept: Window Function

SQL Server query:

```
-- Q42 | Time and Monitoring Lifecycle | Difficult | Window Function
-- Calculate year-over-year change in sites started.

WITH yearly AS (
    SELECT YEAR(start_date) AS start_year, COUNT(*) AS sites_started
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
    GROUP BY YEAR(start_date)
)
SELECT start_year, sites_started,
       LAG(sites_started) OVER (ORDER BY start_year) AS previous_year_sites,
       sites_started - LAG(sites_started) OVER (ORDER BY start_year) AS yoy_change
FROM yearly
ORDER BY start_year;
```

Sample output:

start_year	sites_started	previous_year_sites	yoy_change
1,961.00	2		
1,972.00	1	2	-1
1,973.00	1	1	0
1,975.00	2	1	1
1,977.00	1	2	-1

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps explain how monitoring coverage changed over time and which sites operated for longer periods.

Q43 - Time and Monitoring Lifecycle - Difficult

Business question: Calculate a running total of monitoring sites started over time.

SQL concept: Window Function

SQL Server query:

```
-- Q43 | Time and Monitoring Lifecycle | Difficult | Window Function
-- Calculate a running total of monitoring sites started over time.

WITH yearly AS (
    SELECT YEAR(start_date) AS start_year, COUNT(*) AS sites_started
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
    GROUP BY YEAR(start_date)
)
SELECT start_year, sites_started,
       SUM(sites_started) OVER (ORDER BY start_year ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW) AS
    running_total_sites
FROM yearly
ORDER BY start_year;
```

Sample output:

start_year	sites_started	running_total_sites
1961	2	2
1972	1	3
1973	1	4
1975	2	6
1977	1	7

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps explain how monitoring coverage changed over time and which sites operated for longer periods.

Q44 - Time and Monitoring Lifecycle - Difficult

Business question: Use the lifecycle view to list the longest-operating site in each zone.

SQL concept: VIEW / Window Function

SQL Server query:

```
-- Q44 | Time and Monitoring Lifecycle | Difficult | VIEW / Window Function
-- Use the lifecycle view to list the longest-operating site in each zone.

WITH ranked AS (
    SELECT zone, uk_air_id, site_name, operational_years,
           ROW_NUMBER() OVER (PARTITION BY zone ORDER BY operational_days DESC, site_name) AS rn
    FROM dbo.vw_site_lifecycle_500
)
SELECT zone, uk_air_id, site_name, operational_years
FROM ranked
WHERE rn = 1
ORDER BY zone;
```

Sample output:

zone	uk_air_id	site_name	operational_years
Belfast Metropolitan Urban Area	UKA00175	Belfast East	18.06
Birkenhead Urban Area	UKA13767	ELLESMERE PORT 4N	13.09
Blackpool Urban Area	UKA13448	BLACKPOOL 1N	17
Brighton/Worthing/Littlehampton	UKA00367	Brighton Roadside	12.89
Bristol Urban Area	UKA00262	Bristol Old Market	16.26

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps explain how monitoring coverage changed over time and which sites operated for longer periods.

Q45 - Time and Monitoring Lifecycle - Difficult

Business question: Rank the top 10 longest-operating monitoring sites.

SQL concept: Window Function

SQL Server query:

```
-- Q45 | Time and Monitoring Lifecycle | Difficult | Window Function
-- Rank the top 10 longest-operating monitoring sites.

SELECT TOP (10) uk_air_id, site_name, zone, start_date, end_date, operational_years,
       RANK() OVER (ORDER BY operational_days DESC) AS duration_rank
FROM dbo.vw_site_lifecycle_500
ORDER BY duration_rank, site_name;
```

Sample output:

uk_air_id	site_name	zone	start_date	end_date	operational_years	duration_rank
UKA00002	Cardiff 12	Cardiff Urban Area	1961-04-04	2014-06-09	53.18	1
UKA00001	Bradford 6	West Yorkshire Urban Area	1961-04-04	2009-12-31	48.74	2
UKA00055	Bottesford	East Midlands	1977-10-01	2016-10-17	39.04	3
UKA00153	Billingham	Teesside Urban Area	1987-01-01	2024-06-19	37.46	4
UKA00114	Bannisdale	North West & Merseyside	1986-01-21	2021-12-13	35.89	5

Sample shown: 5 of 10 returned row(s).

Interpretation: The query returns 10 row(s). This helps explain how monitoring coverage changed over time and which sites operated for longer periods.

Identifier, Network and Metadata Coverage

Q46 - Identifier, Network and Metadata Coverage - Easy

Business question: Which sites have an EU site ID available?

SQL concept: Basic WHERE

SQL Server query:

```
-- Q46 | Identifier, Network and Metadata Coverage | Easy | Basic WHERE
-- Which sites have an EU site ID available?

SELECT uk_air_id, eu_site_id, site_name, zone
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
WHERE eu_site_id <> N'Not Available'
ORDER BY zone, site_name;
```

Sample output:

uk_air_id	eu_site_id	site_name	zone
UKA00371	GB0696A	Belfast Clara St	Belfast Metropolitan Urban Area
UKA00175	GB0514A	Belfast East	Belfast Metropolitan Urban Area
UKA00228	GB0798A	Belfast South	Belfast Metropolitan Urban Area
UKA15159	GB0706A	DUNMURRY 3	Belfast Metropolitan Urban Area
UKA00412	GB0727A	Blackpool	Blackpool Urban Area

Sample shown: 5 of 104 returned row(s).

Interpretation: The query returns 104 row(s). This helps check identifier, network and description completeness before relying on the records for reporting.

Q47 - Identifier, Network and Metadata Coverage - Easy

Business question: Which sites have an EMEP site ID available?

SQL concept: Basic WHERE

SQL Server query:

```
-- Q47 | Identifier, Network and Metadata Coverage | Easy | Basic WHERE
-- Which sites have an EMEP site ID available?

SELECT uk_air_id, emep_site_id, site_name, zone
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
WHERE emep_site_id <> N'Not Available'
ORDER BY zone, site_name;
```

Sample output:

uk_air_id	emep_site_id	site_name	zone
UKA00055	GB0032R	Bottesford	East Midlands
UKA00310	GB0047R	Bure Marshes	Eastern
UKA00331	GB0161R	Dennington	Eastern
UKA00102	GB0090R	East Ruston	Eastern
UKA00389	GB0173R	Carradale	Highland

Sample shown: 5 of 13 returned row(s).

Interpretation: The query returns 13 row(s). This helps check identifier, network and description completeness before relying on the records for reporting.

Q48 - Identifier, Network and Metadata Coverage - Easy

Business question: Which sites have both EU and EMEP identifiers available?

SQL concept: Basic WHERE

SQL Server query:

```
-- Q48 | Identifier, Network and Metadata Coverage | Easy | Basic WHERE
-- Which sites have both EU and EMEP identifiers available?

SELECT uk_air_id, eu_site_id, emep_site_id, site_name, zone
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
WHERE eu_site_id <> N'Not Available' AND emep_site_id <> N'Not Available'
ORDER BY zone, site_name;
```

Sample output:

uk_air_id	eu_site_id	emep_site_id	site_name	zone
UKA00055	GB0032R	GB0032R	Bottesford	East Midlands
UKA00310	GB0047R	GB0047R	Bure Marshes	Eastern
UKA00331	GB0161R	GB0161R	Dennington	Eastern

UKA00102	GB0090R	GB0090R	East Ruston	Eastern
UKA00389	GB0173R	GB0173R	Carradale	Highland

Sample shown: 5 of 12 returned row(s).

Interpretation: The query returns 12 row(s). This helps check identifier, network and description completeness before relying on the records for reporting.

Q49 - Identifier, Network and Metadata Coverage - Easy

Business question: Which sites belong to networks containing the word Automatic?

SQL concept: LIKE Filter

SQL Server query:

```
-- Q49 | Identifier, Network and Metadata Coverage | Easy | LIKE Filter
-- Which sites belong to networks containing the word Automatic?

SELECT uk_air_id, site_name, zone, networks
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
WHERE networks LIKE N'%Automatic%'
ORDER BY zone, site_name;
```

Sample output:

uk_air_id	site_name	zone	networks
UKA00371	Belfast Clara St	Belfast Metropolitan Urban Area	Automatic Urban and Rural Monitoring Ne...
UKA00175	Belfast East	Belfast Metropolitan Urban Area	Automatic Urban and Rural Monitoring Ne...
UKA00228	Belfast South	Belfast Metropolitan Urban Area	Automatic Hydrocarbon Network
UKA00412	Blackpool	Blackpool Urban Area	Automatic Urban and Rural Monitoring Ne...
UKA00367	Brighton Roadside	Brighton/Worthing/Littlehampton	Automatic Urban and Rural Monitoring Ne...

Sample shown: 5 of 44 returned row(s).

Interpretation: The query returns 44 row(s). This helps check identifier, network and description completeness before relying on the records for reporting.

Q50 - Identifier, Network and Metadata Coverage - Easy

Business question: Which sites have a site description available?

SQL concept: Basic WHERE

SQL Server query:

```
-- Q50 | Identifier, Network and Metadata Coverage | Easy | Basic WHERE
-- Which sites have a site description available?

SELECT TOP (30) uk_air_id, site_name, zone, LEFT(site_description, 120) AS description_preview
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
WHERE site_description <> N'Not Available'
ORDER BY site_name;
```

Sample output:

uk_air_id	site_name	zone	description_preview
UKA00399	Aberdeen	North East Scotland	The monitoring station is within a self...
UKA00513	Aberdeen Union Street Roadside	North East Scotland	The monitoring station is within a smal...
UKA00949	Ballylinney Church	Northern Ireland	Please note that the location of this s...
UKA00186	Barnsley	Yorkshire & Humberside	Municipal building on edge of a coal mi...
UKA00236	Barnsley 12	Yorkshire & Humberside	The monitoring station is within an exi...

Sample shown: 5 of 30 returned row(s).

Interpretation: The query returns 30 row(s). This helps check identifier, network and description completeness before relying on the records for reporting.

Q51 - Identifier, Network and Metadata Coverage - Medium

Business question: How many records have EU and EMEP identifiers available?

SQL concept: GROUP BY / CASE

SQL Server query:

```
-- Q51 | Identifier, Network and Metadata Coverage | Medium | GROUP BY / CASE
-- How many records have EU and EMEP identifiers available?

SELECT
    CASE WHEN eu_site_id <> N'Not Available' THEN 'EU ID Available' ELSE 'EU ID Missing' END AS eu_id_status,
    CASE WHEN emep_site_id <> N'Not Available' THEN 'EMEP ID Available' ELSE 'EMEP ID Missing' END AS
    emep_id_status,
    COUNT(*) AS site_count
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY CASE WHEN eu_site_id <> N'Not Available' THEN 'EU ID Available' ELSE 'EU ID Missing' END,
         CASE WHEN emep_site_id <> N'Not Available' THEN 'EMEP ID Available' ELSE 'EMEP ID Missing' END
ORDER BY site_count DESC;
```

Sample output:

eu_id_status	emep_id_status	site_count
EU ID Missing	EMEP ID Missing	395
EU ID Available	EMEP ID Missing	92
EU ID Available	EMEP ID Available	12
EU ID Missing	EMEP ID Available	1

Sample shown: 4 of 4 returned row(s).

Interpretation: The query returns 4 row(s). This helps check identifier, network and description completeness before relying on the records for reporting.

Q52 - Identifier, Network and Metadata Coverage - Medium

Business question: What is the EU identifier availability by zone?

SQL concept: GROUP BY / Aggregation

SQL Server query:

```
-- Q52 | Identifier, Network and Metadata Coverage | Medium | GROUP BY / Aggregation
-- What is the EU identifier availability by zone?

SELECT zone,
       SUM(CASE WHEN eu_site_id <> N'Not Available' THEN 1 ELSE 0 END) AS sites_with_eu_id,
       COUNT(*) AS total_sites,
       ROUND(100.0 * SUM(CASE WHEN eu_site_id <> N'Not Available' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
    eu_id_pct
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY zone
ORDER BY eu_id_pct DESC, total_sites DESC;
```

Sample output:

zone	sites_with_eu_id	total_sites	eu_id_pct
Cardiff Urban Area	5	5	100
West Yorkshire Urban Area	2	2	100
Teesside Urban Area	1	1	100
Belfast Metropolitan Urban Area	4	6	66.67
Brighton/Worthing/Littlehampton	2	3	66.67

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps check identifier, network and description completeness before relying on the records for reporting.

Q53 - Identifier, Network and Metadata Coverage - Medium

Business question: What is the EMEP identifier availability by environment type?

SQL concept: GROUP BY / Aggregation

SQL Server query:

```
-- Q53 | Identifier, Network and Metadata Coverage | Medium | GROUP BY / Aggregation
-- What is the EMEP identifier availability by environment type?

SELECT environment_type,
       SUM(CASE WHEN emep_site_id <> N'Not Available' THEN 1 ELSE 0 END) AS sites_with_emep_id,
       COUNT(*) AS total_sites,
```

```
       ROUND(100.0 * SUM(CASE WHEN emep_site_id <> N'Not Available' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
    emep_id_pct
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY environment_type
ORDER BY emep_id_pct DESC, total_sites DESC;
```

Sample output:

environment_type	sites_with_emep_id	total_sites	emep_id_pct
Rural Background	13	58	22.41
Urban Traffic	0	392	0
Urban Background	0	30	0
Suburban	0	10	0
Suburban Background	0	4	0

Sample shown: 5 of 9 returned row(s).

Interpretation: The query returns 9 row(s). This helps check identifier, network and description completeness before relying on the records for reporting.

Q54 - Identifier, Network and Metadata Coverage - Medium

Business question: What is the average available description length by environment type?

SQL concept: GROUP BY / Text Length

SQL Server query:

```
-- Q54 | Identifier, Network and Metadata Coverage | Medium | GROUP BY / Text Length
-- What is the average available description length by environment type?

SELECT environment_type,
       ROUND(AVG(CAST(CASE WHEN site_description <> N'Not Available' THEN LEN(site_description) ELSE NULL END
    AS FLOAT)), 2) AS avg_description_length,
       SUM(CASE WHEN site_description <> N'Not Available' THEN 1 ELSE 0 END) AS descriptions_available
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY environment_type
ORDER BY avg_description_length DESC;
```

Sample output:

environment_type	avg_description_length	descriptions_available
Urban Industrial	416	1
Urban Traffic	286	11
Rural Background	279.5	2
Urban Background	228.75	24
Suburban Background	148.33	3

Sample shown: 5 of 9 returned row(s).

Interpretation: The query returns 9 row(s). This helps check identifier, network and description completeness before relying on the records for reporting.

Q55 - Identifier, Network and Metadata Coverage - Medium

Business question: Which network values appear most often?

SQL concept: GROUP BY / Aggregation

SQL Server query:

```
-- Q55 | Identifier, Network and Metadata Coverage | Medium | GROUP BY / Aggregation
-- Which network values appear most often?

SELECT TOP (15) networks, COUNT(*) AS site_count
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY networks
ORDER BY site_count DESC, networks;
```

Sample output:

networks	site_count
Defra NO2 Diffusion Tube	357
Heavy Metals	31
UKEAP: Precip-Net	30
Automatic Urban and Rural Monitoring Ne...	22
UKEAP: Precip-Net; UKEAP: Rural NO2	10

Sample shown: 5 of 15 returned row(s).

Interpretation: The query returns 15 row(s). This helps check identifier, network and description completeness before relying on the records for reporting.

Q56 - Identifier, Network and Metadata Coverage - Difficult

Business question: Use a CTE to identify weak metadata records by zone.

SQL concept: CTE

SQL Server query:

```
-- Q56 | Identifier, Network and Metadata Coverage | Difficult | CTE
-- Use a CTE to identify weak metadata records by zone.

WITH metadata_flags AS (
    SELECT zone,
           CASE WHEN eu_site_id = N'Not Available' AND emep_site_id = N'Not Available' AND site_description =
    N'Not Available' THEN 1 ELSE 0 END AS weak_metadata_flag
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
)
SELECT zone, SUM(weak_metadata_flag) AS weak_metadata_sites, COUNT(*) AS total_sites,
       ROUND(100.0 * SUM(weak_metadata_flag) / COUNT(*), 2) AS weak_metadata_pct
FROM metadata_flags
GROUP BY zone
ORDER BY weak_metadata_pct DESC, total_sites DESC;
```

Sample output:

zone	weak_metadata_sites	total_sites	weak_metadata_pct
North East	12	12	100
Birkenhead Urban Area	7	7	100
Tyneside	6	6	100
Portsmouth Urban Area	4	4	100
Liverpool Urban Area	2	2	100

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps check identifier, network and description completeness before relying on the records for reporting.

Q57 - Identifier, Network and Metadata Coverage - Difficult

Business question: Rank zones by EU identifier coverage rate.

SQL concept: Window Function

SQL Server query:

```
-- Q57 | Identifier, Network and Metadata Coverage | Difficult | Window Function
-- Rank zones by EU identifier coverage rate.

WITH zone_id_coverage AS (
    SELECT zone, COUNT(*) AS total_sites,
           SUM(CASE WHEN eu_site_id <> N'Not Available' THEN 1 ELSE 0 END) AS sites_with_eu_id
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
    GROUP BY zone
)
SELECT zone, total_sites, sites_with_eu_id,
       ROUND(100.0 * sites_with_eu_id / total_sites, 2) AS eu_id_pct,
       RANK() OVER (ORDER BY 100.0 * sites_with_eu_id / total_sites DESC, total_sites DESC) AS eu_id_rank
FROM zone_id_coverage
ORDER BY eu_id_rank, zone;
```

Sample output:

zone	total_sites	sites_with_eu_id	eu_id_pct	eu_id_rank
Cardiff Urban Area	5	5	100	1
West Yorkshire Urban Area	2	2	100	2
Teesside Urban Area	1	1	100	3
Belfast Metropolitan Urban Area	6	4	66.67	4
Brighton/Worthing/ Littlehampton	3	2	66.67	5

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps check identifier, network and description completeness before relying on the records for reporting.

Q58 - Identifier, Network and Metadata Coverage - Difficult

Business question: Use the identifier quality view to summarize metadata quality bands.

SQL concept: VIEW

SQL Server query:

```
-- Q58 | Identifier, Network and Metadata Coverage | Difficult | VIEW
-- Use the identifier quality view to summarize metadata quality bands.

SELECT metadata_quality_band, COUNT(*) AS site_count
FROM dbo.vw_identifier_quality_500
GROUP BY metadata_quality_band
ORDER BY site_count DESC;
```

Sample output:

metadata_quality_band	site_count
Weak metadata	395
Partial metadata	104
Strong metadata	1

Sample shown: 3 of 3 returned row(s).

Interpretation: The query returns 3 row(s). This helps check identifier, network and description completeness before relying on the records for reporting.

Q59 - Identifier, Network and Metadata Coverage - Difficult

Business question: List the 10 longest site descriptions with row numbers.

SQL concept: Window Function

SQL Server query:

```
-- Q59 | Identifier, Network and Metadata Coverage | Difficult | Window Function
-- List the 10 longest site descriptions with row numbers.

WITH descriptions AS (
    SELECT uk_air_id, site_name, zone, LEN(site_description) AS description_length,
           ROW_NUMBER() OVER (ORDER BY LEN(site_description) DESC, site_name) AS description_rank
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
    WHERE site_description <> N'Not Available'
)
SELECT TOP (10) description_rank, uk_air_id, site_name, zone, description_length
FROM descriptions
ORDER BY description_rank;
```

Sample output:

description_rank	uk_air_id	site_name	zone	description_length
1	UKA00214	Birmingham Centre	West Midlands Urban Area	500
2	UKA00367	Brighton Roadside	Brighton/Worthing/ Littlehampton	500
3	UKA00322	Bury Roadside	Greater Manchester Urban Area	484
4	UKA00236	Barnsley 12	Yorkshire & Humberside	481
5	UKA00262	Bristol Old Market	Bristol Urban Area	437

Sample shown: 5 of 10 returned row(s).

Interpretation: The query returns 10 row(s). This helps check identifier, network and description completeness before relying on the records for reporting.

Q60 - Identifier, Network and Metadata Coverage - Difficult

Business question: Show each network count and its percentage share of the dataset.

SQL concept: Window Function

SQL Server query:

```
-- Q60 | Identifier, Network and Metadata Coverage | Difficult | Window Function
-- Show each network count and its percentage share of the dataset.

WITH network_counts AS (
```

```
    SELECT networks, COUNT(*) AS site_count
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
    GROUP BY networks
)
SELECT networks, site_count,
       ROUND(100.0 * site_count / SUM(site_count) OVER (), 2) AS pct_of_all_sites
FROM network_counts
ORDER BY site_count DESC, networks;
```

Sample output:

networks	site_count	pct_of_all_sites
Defra NO2 Diffusion Tube	357	71.4
Heavy Metals	31	6.2
UKEAP: Precip-Net	30	6
Automatic Urban and Rural Monitoring Ne...	22	4.4
UKEAP: Precip-Net; UKEAP: Rural NO2	10	2

Sample shown: 5 of 28 returned row(s).

Interpretation: The query returns 28 row(s). This helps check identifier, network and description completeness before relying on the records for reporting.

Operational Prioritisation and Data Quality

Q61 - Operational Prioritisation and Data Quality - Easy

Business question: Which traffic or roadside sites have altitude above the dataset average?

SQL concept: Basic WHERE

SQL Server query:

```
-- Q61 | Operational Prioritisation and Data Quality | Easy | Basic WHERE
-- Which traffic or roadside sites have altitude above the dataset average?

SELECT uk_air_id, site_name, zone, environment_type, altitude_m
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
WHERE (environment_type LIKE N'%Traffic%' OR environment_type = N'Roadside')
  AND altitude_m > (SELECT AVG(CAST(altitude_m AS FLOAT)) FROM dbo.ShraddhaYadav_UK_AIR_Sites_500)
ORDER BY altitude_m DESC;
```

Sample output:

uk_air_id	site_name	zone	environment_type	altitude_m
UKA13727	DUNDEE 1N	North East Scotland	Urban Traffic	805
UKA13535	BRYNMAWR 3N	South Wales	Urban Traffic	330
UKA13548	BUXTON DERBYSHIRE 2N	East Midlands	Urban Traffic	296
UKA13664	CROESPENMAEN 3N	South Wales	Urban Traffic	270
UKA13620	CHORLEY 1N	Preston Urban Area	Urban Traffic	268

Sample shown: 5 of 131 returned row(s).

Interpretation: The query returns 131 row(s). This helps prioritise records that need attention based on coverage, site type and data quality.

Q62 - Operational Prioritisation and Data Quality - Easy

Business question: Which 20 sites are furthest west by longitude?

SQL concept: ORDER BY

SQL Server query:

```
-- Q62 | Operational Prioritisation and Data Quality | Easy | ORDER BY
-- Which 20 sites are furthest west by longitude?

SELECT TOP (20) uk_air_id, site_name, zone, longitude
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
ORDER BY longitude ASC, site_name;
```

Sample output:

uk_air_id	site_name	zone	longitude
UKA00343	Derry	Northern Ireland	-7.32911
UKA13692	DERRY 1N	Northern Ireland	-7.31297
UKA13695	DERRY 6N	Northern Ireland	-7.31152
UKA15463	DERRY 11N	Northern Ireland	-7.29599
UKA14553	DERRY 7N	Northern Ireland	-7.29524

Sample shown: 5 of 20 returned row(s).

Interpretation: The query returns 20 row(s). This helps prioritise records that need attention based on coverage, site type and data quality.

Q63 - Operational Prioritisation and Data Quality - Easy

Business question: Which 20 sites are furthest north by latitude?

SQL concept: ORDER BY

SQL Server query:

```
-- Q63 | Operational Prioritisation and Data Quality | Easy | ORDER BY
-- Which 20 sites are furthest north by latitude?

SELECT TOP (20) uk_air_id, site_name, zone, latitude
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
ORDER BY latitude DESC, site_name;
```

Sample output:

uk_air_id	site_name	zone	latitude
UKA00142	Baltasound	Highland	60.7594
UKA00136	Achanarras	Highland	58.4746
UKA00490	Forsinain 2	Highland	58.4119
UKA00300	Forsinain	Highland	58.396
UKA00224	Forsinard	Highland	58.357

Sample shown: 5 of 20 returned row(s).

Interpretation: The query returns 20 row(s). This helps prioritise records that need attention based on coverage, site type and data quality.

Q64 - Operational Prioritisation and Data Quality - Easy

Business question: Which records are part of Automatic monitoring networks?

SQL concept: LIKE Filter

SQL Server query:

```
-- Q64 | Operational Prioritisation and Data Quality | Easy | LIKE Filter
-- Which records are part of Automatic monitoring networks?

SELECT uk_air_id, site_name, zone, networks
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
WHERE networks LIKE N'%Automatic%'
ORDER BY zone, site_name;
```

Sample output:

uk_air_id	site_name	zone	networks
UKA00371	Belfast Clara St	Belfast Metropolitan Urban Area	Automatic Urban and Rural Monitoring Ne...
UKA00175	Belfast East	Belfast Metropolitan Urban Area	Automatic Urban and Rural Monitoring Ne...
UKA00228	Belfast South	Belfast Metropolitan Urban Area	Automatic Hydrocarbon Network
UKA00412	Blackpool	Blackpool Urban Area	Automatic Urban and Rural Monitoring Ne...
UKA00367	Brighton Roadside	Brighton/Worthing/Littlehampton	Automatic Urban and Rural Monitoring Ne...

Sample shown: 5 of 44 returned row(s).

Interpretation: The query returns 44 row(s). This helps prioritise records that need attention based on coverage, site type and data quality.

Q65 - Operational Prioritisation and Data Quality - Easy

Business question: Classify sites into simple altitude bands.

SQL concept: CASE Expression

SQL Server query:

```
-- Q65 | Operational Prioritisation and Data Quality | Easy | CASE Expression
-- Classify sites into simple altitude bands.

SELECT TOP (50) uk_air_id, site_name, zone, altitude_m,
       CASE
           WHEN altitude_m >= 150 THEN 'High altitude'
           WHEN altitude_m >= 50 THEN 'Medium altitude'
           ELSE 'Low altitude'
       END AS altitude_band
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
ORDER BY altitude_m DESC;
```

Sample output:

uk_air_id	site_name	zone	altitude_m	altitude_band
UKA13727	DUNDEE 1N	North East Scotland	805	High altitude
UKA00100	Cow Green Reservoir	North East	510	High altitude
UKA00242	Bylchau	North Wales	390	High altitude
UKA00120	Beddgelert	North Wales	358	High altitude
UKA13535	BRYNMAWR 3N	South Wales	330	High altitude

Sample shown: 5 of 50 returned row(s).

Interpretation: The query returns 50 row(s). This helps prioritise records that need attention based on coverage, site type and data quality.

Q66 - Operational Prioritisation and Data Quality - Medium

Business question: How many sites fall into each metadata quality band?

SQL concept: GROUP BY / CASE

SQL Server query:

```
-- Q66 | Operational Prioritisation and Data Quality | Medium | GROUP BY / CASE
-- How many sites fall into each metadata quality band?

SELECT metadata_quality_band, COUNT(*) AS site_count
FROM dbo.vw_identifier_quality_500
GROUP BY metadata_quality_band
ORDER BY site_count DESC;
```

Sample output:

metadata_quality_band	site_count
Weak metadata	395
Partial metadata	104
Strong metadata	1

Sample shown: 3 of 3 returned row(s).

Interpretation: The query returns 3 row(s). This helps prioritise records that need attention based on coverage, site type and data quality.

Q67 - Operational Prioritisation and Data Quality - Medium

Business question: Which zones have the highest average operational duration?

SQL concept: GROUP BY / Aggregation

SQL Server query:

```
-- Q67 | Operational Prioritisation and Data Quality | Medium | GROUP BY / Aggregation
-- Which zones have the highest average operational duration?

SELECT TOP (15) zone, ROUND(AVG(CAST(operational_days AS FLOAT) / 365.25), 2) AS avg_operational_years,
    COUNT(*) AS site_count
FROM dbo.vw_site_lifecycle_500
GROUP BY zone
ORDER BY avg_operational_years DESC, site_count DESC;
```

Sample output:

zone	avg_operational_years	site_count
Teesside Urban Area	37.46	1
West Yorkshire Urban Area	29.42	2
Cardiff Urban Area	16.24	5
North East Scotland	14.42	9
Blackpool Urban Area	12.03	6

Sample shown: 5 of 15 returned row(s).

Interpretation: The query returns 15 row(s). This helps prioritise records that need attention based on coverage, site type and data quality.

Q68 - Operational Prioritisation and Data Quality - Medium

Business question: What percentage of sites in each zone are traffic or roadside sites?

SQL concept: GROUP BY / CASE

SQL Server query:

```
-- Q68 | Operational Prioritisation and Data Quality | Medium | GROUP BY / CASE
-- What percentage of sites in each zone are traffic or roadside sites?

SELECT zone,
       SUM(CASE WHEN environment_type LIKE N'%Traffic%' OR environment_type = N'Roadside' THEN 1 ELSE 0 END)
    AS traffic_or_roadside_sites,
       COUNT(*) AS total_sites,
       ROUND(100.0 * SUM(CASE WHEN environment_type LIKE N'%Traffic%' OR environment_type = N'Roadside' THEN
    1 ELSE 0 END) / COUNT(*), 2) AS traffic_pct
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY zone
ORDER BY traffic_pct DESC, total_sites DESC;
```

Sample output:

zone	traffic_or_roadside_sites	total_sites	traffic_pct
Birkenhead Urban Area	7	7	100
Tyneside	6	6	100
Portsmouth Urban Area	4	4	100
Brighton/Worthing/Littlehampton	3	3	100
Liverpool Urban Area	2	2	100

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps prioritise records that need attention based on coverage, site type and data quality.

Q69 - Operational Prioritisation and Data Quality - Medium

Business question: Which zones have more than 5 sites and an average altitude above 50 metres?

SQL concept: GROUP BY / HAVING

SQL Server query:

```
-- Q69 | Operational Prioritisation and Data Quality | Medium | GROUP BY / HAVING
-- Which zones have more than 5 sites and an average altitude above 50 metres?

SELECT zone, COUNT(*) AS total_sites, ROUND(AVG(CAST(altitude_m AS FLOAT)), 2) AS avg_altitude_m
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY zone
HAVING COUNT(*) > 5 AND AVG(CAST(altitude_m AS FLOAT)) > 50
ORDER BY avg_altitude_m DESC, total_sites DESC;
```

Sample output:

zone	total_sites	avg_altitude_m
North East Scotland	9	157.67
West Midlands Urban Area	21	133.19
Greater Manchester Urban Area	6	105.33
West Midlands	28	101.36
North East	12	98.42

Sample shown: 5 of 12 returned row(s).

Interpretation: The query returns 12 row(s). This helps prioritise records that need attention based on coverage, site type and data quality.

Q70 - Operational Prioritisation and Data Quality - Medium

Business question: Summarize description availability by zone.

SQL concept: GROUP BY / CASE

SQL Server query:

```
-- Q70 | Operational Prioritisation and Data Quality | Medium | GROUP BY / CASE
-- Summarize description availability by zone.

SELECT zone,
       SUM(CASE WHEN site_description <> N'Not Available' THEN 1 ELSE 0 END) AS descriptions_available,
       COUNT(*) AS total_sites,
       ROUND(100.0 * SUM(CASE WHEN site_description <> N'Not Available' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
    description_available_pct
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
GROUP BY zone
ORDER BY description_available_pct DESC, total_sites DESC;
```

Sample output:

zone	descriptions_available	total_sites	description_available_pct
Teesside Urban Area	1	1	100
Belfast Metropolitan Urban Area	3	6	50
Coventry/Bedworth	2	4	50
West Yorkshire Urban Area	1	2	50
Cardiff Urban Area	2	5	40

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps prioritise records that need attention based on coverage, site type and data quality.

Q71 - Operational Prioritisation and Data Quality - Difficult

Business question: Place sites into altitude quartiles using NTILE.

SQL concept: Window Function

SQL Server query:

```
-- Q71 | Operational Prioritisation and Data Quality | Difficult | Window Function
-- Place sites into altitude quartiles using NTILE.

SELECT uk_air_id, site_name, zone, environment_type, altitude_m,
       NTILE(4) OVER (ORDER BY altitude_m) AS altitude_quartile
FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
ORDER BY altitude_quartile, altitude_m;
```

Sample output:

uk_air_id	site_name	zone	environment_type	altitude_m	altitude_quartile
UKA15383	BALLYNAHINCH 8N	Northern Ireland	Urban Traffic	0	1
UKA15434	CAERPHILLY 8N	South Wales	Urban Traffic	0	1
UKA15449	CLYDEBANK 5N	Central Scotland	Urban Traffic	0	1
UKA13717	DOWNPATRICK 2N	Northern Ireland	Urban Traffic	1	1
UKA00126	Avonmouth 1	Bristol Urban Area	Urban Traffic	2	1

Sample shown: 5 of 500 returned row(s).

Interpretation: The query returns 500 row(s). This helps prioritise records that need attention based on coverage, site type and data quality.

Q72 - Operational Prioritisation and Data Quality - Difficult

Business question: Classify zones as high, medium, or low coverage compared with the dataset average.

SQL concept: CTE

SQL Server query:

```
-- Q72 | Operational Prioritisation and Data Quality | Difficult | CTE
-- Classify zones as high, medium, or low coverage compared with the dataset average.

WITH zone_counts AS (
    SELECT zone, COUNT(*) AS total_sites
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
    GROUP BY zone
), stats AS (
    SELECT AVG(CAST(total_sites AS FLOAT)) AS avg_zone_sites
    FROM zone_counts
)
SELECT z.zone, z.total_sites, ROUND(s.avg_zone_sites, 2) AS avg_zone_sites,
```

```
        CASE
            WHEN z.total_sites >= s.avg_zone_sites * 1.5 THEN 'High coverage zone'
            WHEN z.total_sites >= s.avg_zone_sites THEN 'Medium coverage zone'
            ELSE 'Low coverage zone'
        END AS coverage_category
FROM zone_counts z
CROSS JOIN stats s
ORDER BY z.total_sites DESC, z.zone;
```

Sample output:

zone	total_sites	avg_zone_sites	coverage_category
South East	56	14.29	High coverage zone
Northern Ireland	44	14.29	High coverage zone
Eastern	40	14.29	High coverage zone
South Wales	38	14.29	High coverage zone
East Midlands	34	14.29	High coverage zone

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps prioritise records that need attention based on coverage, site type and data quality.

Q73 - Operational Prioritisation and Data Quality - Difficult

Business question: Create a data readiness summary combining identifier and description status.

SQL concept: CTE / CASE

SQL Server query:

```
-- Q73 | Operational Prioritisation and Data Quality | Difficult | CTE / CASE
-- Create a data readiness summary combining identifier and description status.

WITH readiness AS (
    SELECT
        CASE WHEN eu_site_id <> N'Not Available' OR emep_site_id <> N'Not Available' THEN 'Has external
    identifier' ELSE 'No external identifier' END AS identifier_status,
        CASE WHEN site_description <> N'Not Available' THEN 'Description available' ELSE 'Description
    missing' END AS description_status
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
)
SELECT identifier_status, description_status, COUNT(*) AS site_count
FROM readiness
GROUP BY identifier_status, description_status
ORDER BY identifier_status, description_status;
```

Sample output:

identifier_status	description_status	site_count
Has external identifier	Description available	41
Has external identifier	Description missing	64
No external identifier	Description missing	395

Sample shown: 3 of 3 returned row(s).

Interpretation: The query returns 3 row(s). This helps prioritise records that need attention based on coverage, site type and data quality.

Q74 - Operational Prioritisation and Data Quality - Difficult

Business question: Calculate a 3-year moving average of sites started by year.

SQL concept: Window Function

SQL Server query:

```
-- Q74 | Operational Prioritisation and Data Quality | Difficult | Window Function
-- Calculate a 3-year moving average of sites started by year.

WITH yearly AS (
    SELECT YEAR(start_date) AS start_year, COUNT(*) AS sites_started
    FROM dbo.ShraddhaYadav_UK_AIR_Sites_500
    GROUP BY YEAR(start_date)
)
SELECT start_year, sites_started,
       ROUND(AVG(CAST(sites_started AS FLOAT)) OVER (ORDER BY start_year ROWS BETWEEN 2 PRECEDING AND CURRENT
    ROW), 2) AS moving_avg_3_years
```

```
FROM yearly
ORDER BY start_year;
```

Sample output:

start_year	sites_started	moving_avg_3_years
1,961.00	2	2
1,972.00	1	1.5
1,973.00	1	1.33
1,975.00	2	1.33
1,977.00	1	1.33

Sample shown: 5 of 35 returned row(s).

Interpretation: The query returns 35 row(s). This helps prioritise records that need attention based on coverage, site type and data quality.

Q75 - Operational Prioritisation and Data Quality - Difficult

Business question: Create a final prioritisation list for traffic or roadside sites with weak or partial metadata.

SQL concept: CTE / Window Function

SQL Server query:

```
-- Q75 | Operational Prioritisation and Data Quality | Difficult | CTE / Window Function
-- Create a final prioritisation list for traffic or roadside sites with weak or partial metadata.

WITH candidate_sites AS (
    SELECT q.uk_air_id, q.site_name, q.zone, q.environment_type, q.metadata_quality_band,
           l.operational_years,
           CASE WHEN q.metadata_quality_band = 'Weak metadata' THEN 3
                WHEN q.metadata_quality_band = 'Partial metadata' THEN 2
                ELSE 1 END AS metadata_priority_score
    FROM dbo.vw_identifier_quality_500 q
    INNER JOIN dbo.vw_site_lifecycle_500 l ON q.id = l.id
    WHERE q.environment_type LIKE N'%Traffic%' OR q.environment_type = N'Roadside'
), ranked AS (
    SELECT *,
           RANK() OVER (ORDER BY metadata_priority_score DESC, operational_years DESC, site_name) AS
    priority_rank
    FROM candidate_sites
)
SELECT TOP (25) priority_rank, uk_air_id, site_name, zone, environment_type, metadata_quality_band,
    operational_years
FROM ranked
ORDER BY priority_rank, site_name;
```

Sample output:

priority_rank	uk_air_id	site_name	zone	environment_type	metadata_quality_band	operational_years
1	UKA13447	BISLEY 1N	South East	Urban Traffic	Weak metadata	17.92
2	UKA13726	DUMBARTON 7N	Central Scotland	Urban Traffic	Weak metadata	17.58
3	UKA13486	BRENT 41N	Greater London Urban Area	Urban Traffic	Weak metadata	17.5
4	UKA13552	CAERPHILLY 7N	South Wales	Urban Traffic	Weak metadata	17.06
5	UKA13664	CROESPENMAEN 3N	South Wales	Urban Traffic	Weak metadata	17.06

Sample shown: 5 of 25 returned row(s).

Interpretation: The query returns 25 row(s). This helps prioritise records that need attention based on coverage, site type and data quality.

7. Closing Reflection

The biggest challenge was making sure the queries matched the actual columns and used SQL Server syntax correctly. The dataset uses many text identifiers and long descriptive fields, so the data types had to be chosen carefully to avoid truncation. The most useful insight was that monitoring coverage is not only a count of sites; it also depends on zone coverage, environment type, network coverage, site duration and metadata quality. This is why the project uses five perspectives rather than one simple summary. The 500-row sample shows how a local authority manager can use SQL to move from raw monitoring records to practical coverage and data-quality checks.

Appendix A - SQL Server Data Dictionary

Column	Recommended SQL Server Data Type	Reason	Check / Evidence
id	INT NOT NULL PRIMARY KEY	Unique row identifier; numeric whole nu...	Example: 1
uk_air_id	NVARCHAR(20) NOT NULL	UK-AIR site identifier. Stored as text ...	Max length: 8
eu_site_id	NVARCHAR(50) NOT NULL	EU identifier where available; Not Avai...	Max length: 13
emep_site_id	NVARCHAR(50) NOT NULL	EMEP identifier where available; Not Av...	Max length: 13
site_name	NVARCHAR(255) NOT NULL	Monitoring site name; text field with p...	Max length: 41
environment_type	NVARCHAR(100) NOT NULL	Site character such as Urban Traffic, U...	9 distinct values
zone	NVARCHAR(150) NOT NULL	UK zone or urban area used for geograph...	35 distinct values
start_date	DATE NOT NULL	Monitoring site start date in yyyy-mm-d...	Range: 1961-04-04 to 2022-04-01

Sample shown: 8 of 8 returned row(s).

Column	Recommended SQL Server Data Type	Reason	Check / Evidence
end_date	DATE NOT NULL	Monitoring site end date in yyyy-mm-dd ...	Range: 1975-10-09 to 2025-01-09
latitude	DECIMAL(10,6) NOT NULL	Latitude coordinate for mapping; decima...	Six decimal places retained
longitude	DECIMAL(10,6) NOT NULL	Longitude coordinate for mapping; decim...	Six decimal places retained
northing	DECIMAL(12,2) NOT NULL	British National Grid northing; numeric...	Two decimals retained
easting	DECIMAL(12,2) NOT NULL	British National Grid easting; numeric ...	Two decimals retained
altitude_m	DECIMAL(10,2) NOT NULL	Altitude in metres; numeric decimal use...	Two decimals retained
networks	NVARCHAR(MAX) NOT NULL	Network names can be long and can conta...	Max length: 186
site_description	NVARCHAR(MAX) NOT NULL	Long site description text can exceed n...	Max length: 500

Sample shown: 8 of 8 returned row(s).